AI Smart Navigation Assistant for Visually Impaired People Using Raspberry Pi

Smart Cane Environmental Awareness System

IoT Term Project · First Draft Presentation

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A. Main Idea & Objective

Problem: Visually impaired users struggle to navigate indoor environments safely and independently, especially in unfamiliar spaces.

Original vision — Visual SLAM
Full indoor mapping using ORB-SLAM3 and ROS, building a 3D map in real time and navigating autonomously.

➡️ Deprioritized: requires IMU, significant compute, and complex calibration beyond current scope.

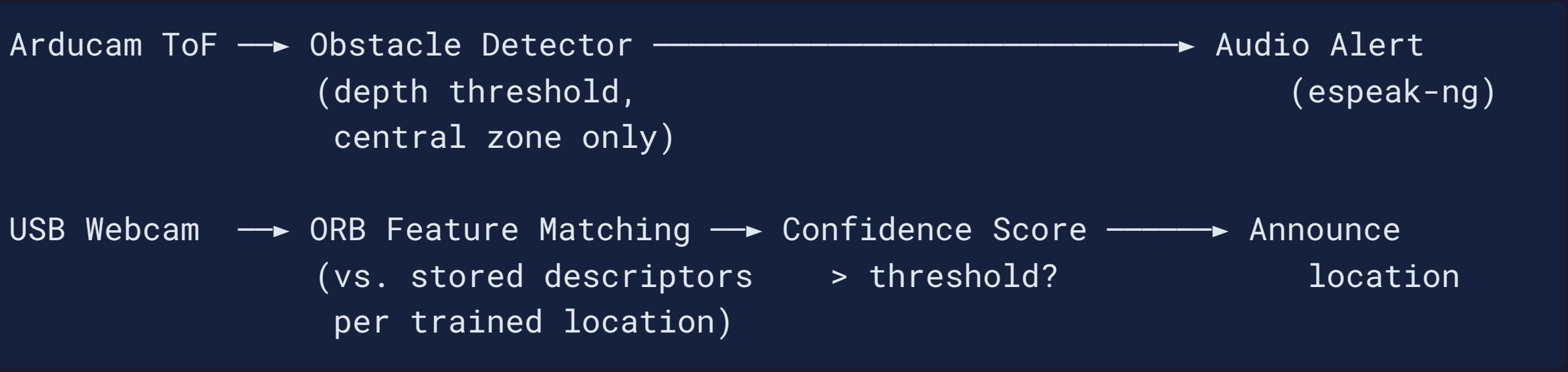
Revised scope — Awareness System
Two focused functions that are achievable, reliable, and genuinely useful:

- 🔴 **Obstacle detection** via ToF depth camera
- 🔵 **Location recognition** via webcam + ORB features
- 🔊 **Audio feedback** via Bluetooth headphones

> *"Obstacle 0.8m ahead."* *"You are in the office."*

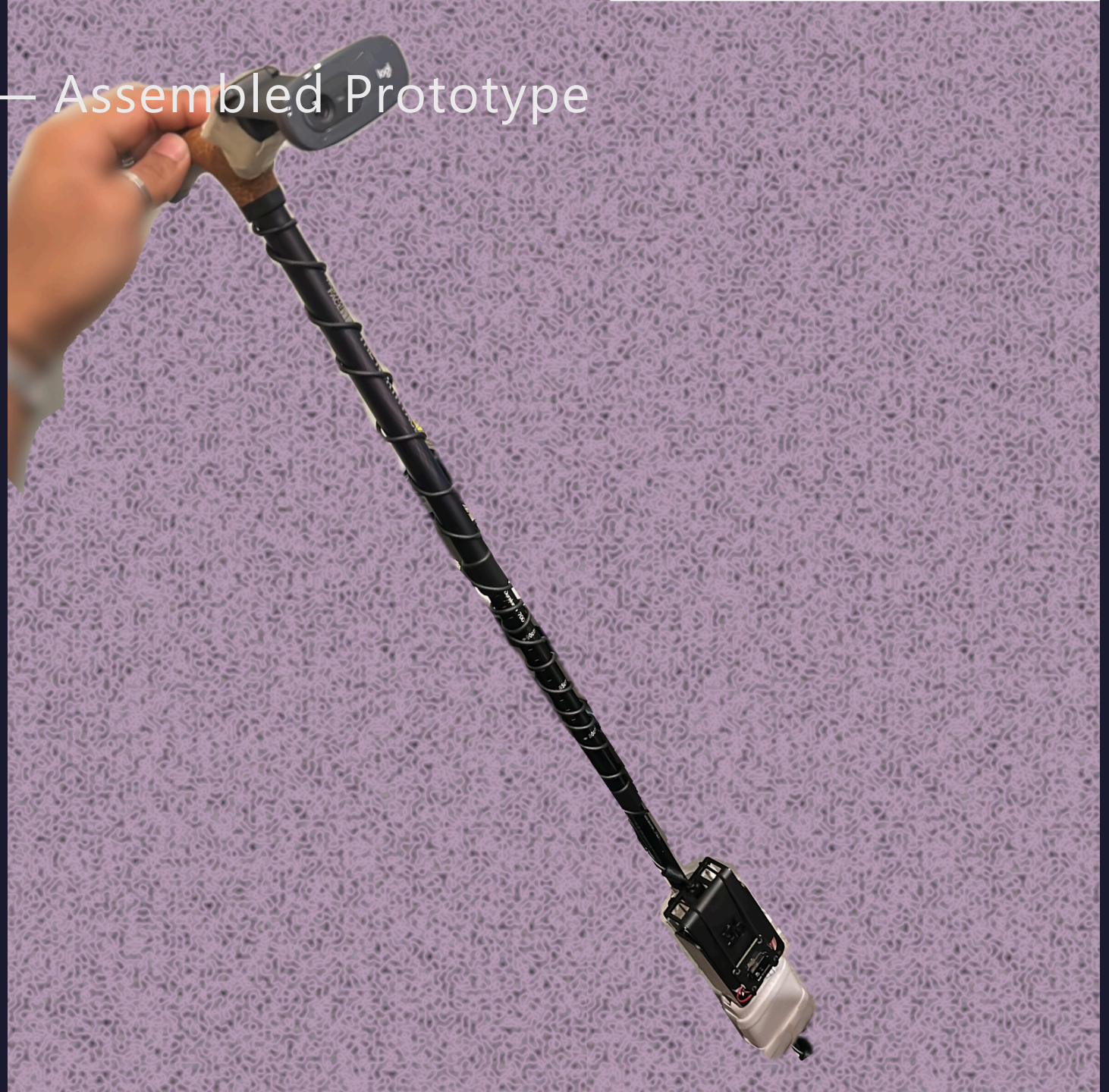
B. System Design & Methodology

Awareness loop at 2 Hz — fully offline, no cloud:



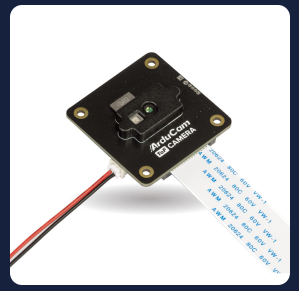
	What	How	When
Obstacle	Depth < threshold in central zone	ToF frame	Every cycle
Location	ORB descriptors matched vs. training	Webcam frame	Every 3rd cycle
Audio	espeak-ng subprocess	Bluetooth headphones	On event

D. Hardware — Assembled Prototype



D. Hardware Components

Raspberry Pi 4
Central compute
4GB RAM



Arducam ToF B0410
Depth sensing
CSI · up to 4m

Logitech C270
Location recognition
USB · 720p

BT Headphones
Audio output
A2DP · offline TTS

Component	Key Spec	Role
Raspberry Pi 4 (4GB)	ARM Cortex-A72 · 4GB LPDDR4	Runs all processing locally
Arducam ToF B0410	240×180 · 10 FPS · 4m range	Forward obstacle depth
Logitech C270	640×480 · 30 FPS · UVC	ORB-based room recognition
Bluetooth Headphones	A2DP audio	espeak-ng TTS output

C. Code — Obstacle Detection

src/nav_assistant/obstacle/detector.py

```python
def check(self, tof_frame: ToFFrame) -> Optional[ObstacleAlert]:
    # Check only the central third of the frame (forward path)
    min_depth = tof_frame.forward_min_depth(zone_fraction=0.33)

    if min_depth > 0 and min_depth < self._threshold:
        now = time.monotonic()
        # Cooldown prevents repeated alerts for the same obstacle
        if now - self._last_alert_time >= self._cooldown:
            self._last_alert_time = now
            return ObstacleAlert(min_depth=min_depth, ...)

    return None
```

- Central **33% of the depth frame** = forward path only

Output examples:

C. Code — Location Recognition

src/nav_assistant/localization/place_recognizer.py

```python
def recognize(self, gray_frame: np.ndarray) -> RecognitionResult:
    _, query_descs = self._orb.detectAndCompute(gray_frame, None)

    for waypoint, stored_descs in self._index:
        matches = self._matcher.knnMatch(query_descs, stored_descs, k=2)
        # Lowe's ratio test — keep only unambiguous matches
        good = [m for m, n in matches if m.distance < 0.75 * n.distance]
        if len(good) > best_good:
            best_good, best_wp = len(good), waypoint

    confidence = min(1.0, best_good / 250)
    return RecognitionResult(waypoint=best_wp, confidence=confidence, ...)
```

Training data per location

- 2-minute walk-around session
- 1 frame every 3 seconds → 40 frames

Current Progress

Phase	Status	What was done
1 — Sensor Integration	✅ Complete	Webcam + ToF validated on RPi hardware
2 — Location Training	✅ Complete	2 locations, 5000 descriptors each
3 — Awareness Loop	🔄 In Progress	Loop built and running, testing ongoing
4 — Tuning & Polish	⏳ Pending	Confidence and threshold tuning

Scripts running on hardware today:

Script	Purpose
`test_sensors.py`	Validates webcam + ToF connectivity
`train.py`	2-min capture sessions, auto-detects cameras
`awareness.py`	Real-time obstacle + location feedback loop

E. Challenges & Solutions

Challenge	Solution
Webcam gets random `/dev/videoX` index across reboots	Parse `v4l2-ctl --list-devices` at startup, match by device name
Arducam SDK `DeviceType.TOF` doesn't exist in installed version	Use `FrameType.DEPTH` — discovered by reading official Python examples
`pyttsx3` crashes on Python 3.13 + espeak-ng	Call `espeak-ng` directly via `subprocess` — no TTS library needed
Webcam returns blank frames right after `open()`	Discard first 5 frames to allow UVC sensor warmup
Single training frame insufficient for reliable recognition	2-minute continuous capture session (40 diverse frames per location)

F. Expected Output & Benefits

What the user hears:

```
"Awareness system started."

"You are in the office."

"Obstacle ahead, 0.8 metres."

"You are in the hallway."

"Warning! Obstacle very close."
```

Benefits:

- **Offline** — works without internet in any indoor space
- **Low cost** — ~€80 total hardware
- **Real-time** — 2 Hz loop, < 500ms response
- **Wearable** — Bluetooth audio, cane form factor
- **Extensible** — architecture ready for navigation features

G. Next Steps & Future Plan

This week

- End-to-end test of `awareness.py` on hardware
- Tune `confidence_threshold` and `alert_threshold_m`
- Train more locations in real environment

Before submission

- Startup/shutdown audio announcements
- CPU profiling and frame rate optimization
- Final demo recording

Future work (post-submission)

Originally planned as v1 — preserved as future roadmap:

- 🗺️ Visual SLAM / indoor map building
- 🧭 Route planning (Dijkstra on waypoint graph)
- 🦾 Turn-by-turn navigation instructions
- 📐 IMU dead-reckoning between locations
- 🤖 MobileNetV3 for richer scene recognition

Thank You

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`github.com/w1ggor/smart-nav-cane`

Questions?